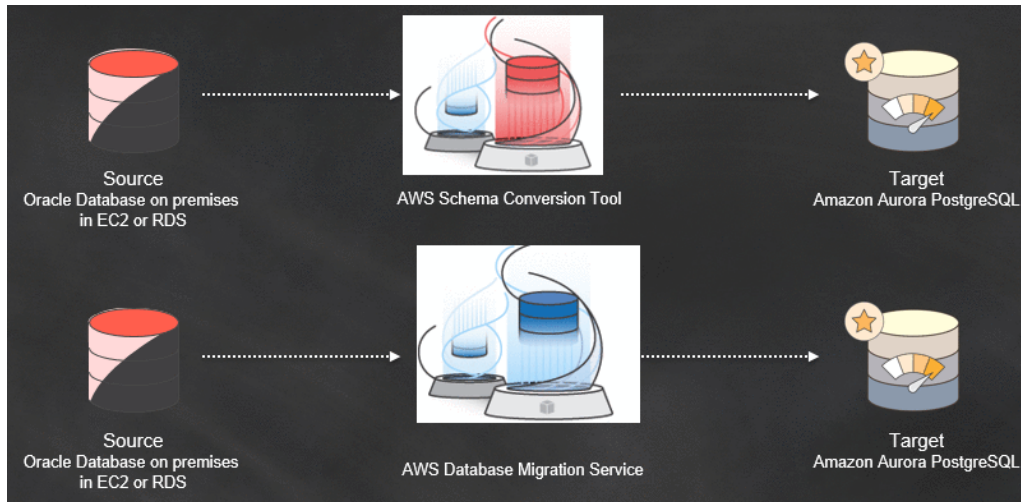


PSP Aurora Postgres Database Migration

- Migration flow
- High Level Plan
- Open Issues
- AWS DataLabs

Migration flow



High Level Plan

[illegible]

8	H/A DR Upgrade	YET TO START	Milestone8											
9	Final Migration - Tip-Toes stabilize	YET TO START	Milestone9											
10	Cutover	YET TO START	Milestone10											

Open Issues

Area	Issue	Severity	Comments	AWS Case#	JIRA Link	Status
#1 Partitioning	- check functional and performance impact of changing partitioning key - Cannot have NOT VALIDATED FK constraint on partitioned tables	P0	PSP Data migration			Under Investigation
Data Migration	- Issues while migrating data - DDL replication	P0	Need to test at production scale			in progress
Data Comparison	- check for CLOB data type comparison - Got OOS on multiple tables. - Check to compare partitioned tables data between oracle and Postgres	P0	Need to test at production scale			Pending

AWS DataLabs

Category	Task	Comments
Prep-work	☒ migrate monolith prod on APG 12 ☐ Set up a perf test suite ☐ Write query templates for Top N queries to assess performance ☐ Top N by elapsed time ☐ Top N by IO time ☐ Top N by CPU time ☐ Top N by executions ☐ Come up with the priority items for DataLabs, prepare a scope document with expected outcome	
During DataLabs	Solution for date range partitioning	

Milestone 1

Objectives:

- Convert schema & postgres optimal data type from Oracle to Postgres to get familiar with AWS SCT, DMS tools & identify issues
- Familiarize with Aurora Postgres

Task Description	Status	Details	Comments
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1	Prep work and Schema Conversion for Postgres ☒ Data type mapping ☒ Tables ☒ Regular Indexes ☒ Function Based Indexes ☒ Primary Keys ☒ Foreign Keys ☒ Triggers ☐ Packages, Procedures & Functions ☒ Partitioning	BLOCKER	Generated DDL via SCT and changed data types accordingly. This is a short term solution until schema generator is updated to support Postgres.	This needs to be generated from application.
2	Database Configuration ☒ DB Timezone in Pacific ☒ Set session timezone ☐ Set Lock timeouts ☒ DB setup on Aurora Postgres ☐ Users & Permissions ☐ Security, Audit	GREEN	Able to setup Aurora PostgreSQL cluster via IPS including wavefront monitoring. While also working with IPS team for any bugs/issues/features.	

Milestone 2

Objectives:

- Replicate data from Oracle to Postgres and vice-versa (if required).

	Task Description	Status	Details	Comments
1	Tools evaluated/ to be evaluated	YELLOW	☐ Data Migration ☒ DMS	Data migration and Data parity tool to be evaluated at production grade.
2	Primary/Monolith Database	WORK IN PROGRESS	Data Migration from Oracle to Aurora ☐ Initial sync – POC in progress ☐ CDC/ongoing data migration ☐ Improve DMS replication performance ☐ DDL Replication	
3	Secondary/IBOB Database	GREEN	Data Migration from Oracle to Aurora ☒ Initial synch – Tested with latest 3 partitions ☒ CDC/ongoing data migration ☒ Improve DMS replication performance	
4	Replication from Postgres to Oracle	NA	Currently assuming Application will do DUAL write. Need to confirm on final migration strategy	

Milestone 3

Objectives:

- Validate data between Oracle and Postgres to certify for switchover and rollback.

	Task Description	Status	Details	Comments
1	Data Validation - DMS - Full sync testing - Incremental sync testing	NOT SCALING	☐ Evaluate DMS for data validation ☒ Compare non-partition tables without CLOB columns ☐ DMS validation scalability for busy tables. ☐ Can't compare partition tables ☐ CLOB columns appearing as OOS most of time	
2	Data Validation - DB SOLO		Yet to start	

Milestone 4

Objectives:

- Data Model - Identify scope, test & implement optimization across various areas for Postgres

	Task Description	Status	Comments
1	Physical data model optimization for Postgres ☐ Indexing ☐ Partitioning ☐ Constraints ☐ Multiple schemas/databases per Aurora cluster		
2			

Milestone 5

- Performance -

	Task Description	Status	Comments
1	☐ DB Server Performance ☐ SQL Performance ☐ SQL Plan Management ☐ Workload Repository		
2	Manual Query Rewrite for testing ☐ Top N by elapsed time ☐ Top N by IO time ☐ Top N by CPU time ☐ Top N by executions		

Milestone 6

- Functional and Regression Testing

	Task Description	Status	Comments
1	☐ Functionality testing ☐ Perf Regression Test using application		
2			

Milestone 7

- Monitoring

	Task Description	Status	Comments
1	☐ DB Monitoring ☐ DB Alerts & Pager ☐ DB Dashboards		IPS
2	Jobs/Scripts/Utilities		
3	Data to Downstream • IAC data ingestion		IPS

Milestone 8

- DR and High Availability

	Task Description	Status	Comments
1	☐ Disaster Recovery plan for Aurora ☐ High Availability Testing ☐ Major version upgrade for Postgres		
2			

Milestone 9

Objective:

- Tip-toes and Switchover

	Task Description	Status	Comments
1	☐ Switchover/switchback process for tip-toe ☐ Tip-toes & Stabilize		
2			

Milestone 10

- TouchDown

	Task Description	Status	Comments
1	☐ Prepare plan for cutover ☐ Cutover/Touch Down		
2			